Sujin Jin

Computer Vision · Large Language Models · 3D Reconstruction for Construction & Infrastructure Safety

Suwon, Republic of Korea | +82-10-2211-8922 | kane0113@skku.edu | sujin1229.github.io | github.com/sujin1229

SUMMARY

AI-driven civil infrastructure researcher focused on computer vision, large language models, multimodal sensing, and 3D reconstruction for smart construction, structural health monitoring, and infrastructure maintenance. My work develops explainable, data-driven frameworks that integrate visual, nondestructive testing (NDT), BIM, LiDAR, and ontology-based information to support automated defect diagnosis, safety monitoring, and digital-twin applications. I am particularly interested in advancing multimodal AI and LLM-based reasoning systems that interpret complex engineering data and translate it into actionable insight for civil infrastructure — a field-first approach that returns technology to the practitioners on site.

EDUCATION

M.S., Global Smart City — Sungkyunkwan University (SKKU), Suwon Sep 2025 – Present

Smart Construction IT (SCIT) Lab · Advisor: Prof. Seunghee Park · **GPA 4.5 / 4.5**

Thesis: A Virtual Model for Predicting AHU Return-Water Temperature for Energy-Optimal HVAC Operation.

Coursework: AI & Deep Learning in Construction Data Analysis · Engineering GPT-Based Multi-Agent Systems · Machine Learning in Construction Informatics

B.Eng., Civil, Architectural Engineering & Landscape Architecture — SKKU, Suwon Mar 2022 – Aug 2025

Summa cum laude · GPA 4.4 / 4.5 (Major 4.37 / 4.5)

PUBLICATIONS

- **Jin, S.**, Song, H., Kang, J., Yu, B., & Park, S. (2026). Concrete crack reasoning: Explainable defect diagnosis incorporating generative pretrained transformer-4 and multimodal nondestructive testing data. *Automation in Construction, 181*, 106661. doi:10.1016/j.autcon.2025.106661
- Ko, D., Park, M., **Jin, S.**, Aung, P. P. W., & Park, S. (2025). Vision-based automated cable tension monitoring using pixel tracking. *Automation in Construction, 179*, 106488. doi:10.1016/j.autcon.2025.106488

RESEARCH EXPERIENCE

AI-Agent-Based Automated HOS Analysis System · URP Project 2024

- Developed an LLM multi-agent system that autonomously performs Holistic Operational Signature (HOS) analysis, moving beyond image-based interpretation of prior work.

- Designed a Brick Schema ontology–driven LLM to extract key variables and generate feature values, and optimized k-means cluster counts via LLM-based knowledge engineering.
- Reached **99.53% cluster-classification accuracy (2 errors / 2,541 cases)** using a feature-value transformation that overcomes limits of image-based LLMs (e.g., GPT-4o).

NeRF to BIM: Integration of LiDAR & Neural Radiance Fields 2024

- Compared NeRF and LiDAR data against BIM models to evaluate NeRF for precise 3D reconstruction of construction data.
- Researched automated BIM data transformation and 3D visualization using digital-twin concepts.

AI-Based Worker Safety Monitoring System for Construction Sites 2024

- Designed an AI system for real-time monitoring of construction-worker safety, fusing biometric and environmental data to preemptively detect risk factors.
- Integrated AI safety monitoring with Structural Health Monitoring (SHM) to build a site safety-evaluation model.

Eco-Friendly Water-Quality Improvement System Using Wind-Powered Ventilation 2024

- Designed a purification system driven by wind-powered natural ventilation and validated water-quality gains across air-injection methods.

Modular Urban Tensegrity Pedestrian Overpass Design 2023

- Designed and visualized a modular tensegrity pedestrian overpass, establishing a BIM-based process with Twinmotion 3D visualization.
- Studied modular construction methods to cut construction time and improve maintenance efficiency.

Olympic Expressway Undergrounding Project 2023

- Conducted feasibility studies and master planning for undergrounding the Olympic Expressway; derived the optimal design through economic analysis, simulation, and BIM modeling.

Bambat Overpass Undergrounding & Environmental Improvement Project 2023

- Researched a BIM-based redevelopment plan to underground an existing overpass, analyzing urban environmental impact and economic feasibility.

CONFERENCE PRESENTATIONS

Drone-Based 3D Mapping and Video Data Fusion for Spatially Intelligent Safety Monitoring 2025

Computational Structural Engineering Institute of Korea (COSEIK) · Oral Presentation · **Excellent Paper Award**

An Explainable Diagnostic Framework for Concrete Defects Using Multimodal NDT and GPT-4 2025
Korea Institute for Structural Maintenance and Inspection (KSMI) · Oral Presentation

Multi-modal Deep Learning-Based Bridge Deterioration Detection using UT and GPR 2025
Korea Society of Hazard Mitigation (KOSHAM) · Oral Presentation

PROFESSIONAL EXPERIENCE & ASSISTANTSHIPS

Research Assistant — Smart Construction IT (SCIT) Lab, SKKU Jun – Sep 2024
Computer vision, infrastructure maintenance, and structural health monitoring. Advisor: Prof. Seunghee Park.

Teaching Assistant — Creative Convergence Design, SKKU Aug – Dec 2025
Guided first-year students in applying engineering thinking to real-world design problems; facilitated design projects and discussions. Supervisor: Prof. Jiyoo Park.

AWARDS & HONORS

Dec 2024 **National Scholarship for Outstanding Students in Science & Engineering** — Minister of Science and ICT

Nov 2025 **Excellent Paper Award** — Computational Structural Engineering Institute of Korea

Oct 2025 **Next-Generation Engineer Award (Excellence)** — Korea Institute for the Promotion of Engineering Education (IPEK)

Aug 2025 **SKKU Graduate Scholarship (Full Tuition)** — Sungkyunkwan University

Sep 2024 **Korea Engineering Industry Competition — 4th Place** — Korea Engineering & Consulting Association

Nov 2023 & 2024 **KIBIM BIM Competition — Encouragement Award** — Korean Institute of Building Information Modeling

Sep 2023 **Korea Engineering Industry Competition — Excellence Award** — Korea Engineering & Consulting Association

Sep 2023 **8th National University CM Competition — 4th Place** — MOOYOUNG Construction Management

SKILLS

Programming & ML Python, HTML, Streamlit, LangChain (LangSmith, LangGraph), LLM engineering (GPT, Llama), AI engineering

CV & 3D PyTorch, YOLO, VGGT, COLMAP, Qwen2.5-VL, 3D reconstruction

Robotics ROS 2, Gazebo, Nav2

BIM & CAD AutoCAD, Revit, ALLPLAN, SketchUp, Twinmotion

Data & Ontology Tableau, Brick Schema

Languages Korean (native), English (TOEFL iBT)

EXTRACURRICULAR ACTIVITIES & CERTIFICATES

Member, ACT:U:ALL BIM Construction Study Club — SKKU Mar 2023 – Aug 2025
Student-led study club on BIM-based construction; seminars and team projects.

KIC DC Tech Frontier Program — KIC Washington, with George Washington University · Washington, D.C. Jul 2024

Certificates — POSCO E&C Smart Construction Academy; Entrepreneurship Exploration Program (Ministry of Science and ICT).